

## EDUCATION

---

**University of California, Irvine** 2020 - Present  
*Ph.D. candidate in the Department of Physics and Astronomy*  
*Member of the ATLAS collaboration*  
*Advisor: Daniel Whiteson*

**University of Notre Dame** 2016 - 2020  
*Bachelor of Science, Magna Cum Laude*  
*Majors: Physics and Program of Liberal Studies*

## RESEARCH EXPERIENCE

---

**Machine learning based unfolding** Ongoing  
*- Develop novel unfolding methods based on generative machine learning models, and contribute to community comparison studies*  
*- Study the usage of un-binned and highly differential cross section measurements for parameter estimation*  
*- Deploy machine learning based unfolding to perform a hyper-differential cross section measurement of Z+jets production with the ATLAS detector*

**Constituent based top tagging** 2022 - 2024  
*- Bench-marked several constituent top-tagging algorithms on realistic detector simulation*  
*- Assessed systematic uncertainties in tagger performance*  
*- Prepared research grade open data records for use by the broader particle physics community*

**Calibration of the ATLAS New Small Wheel** 2022 - 2023  
*- Calibrated the peak detector output (PDO) readout used in the ATLAS New Small Wheel muon detector*  
*- Participated in operations of the detector during Run 3*

**Physics inspired neural networks** 2018 - 2020  
*- Deploy a physically motivated neural network architecture on a classification problem in particle physics*

**CMS track trigger development** 2017 - 2019  
*- Improved the efficiency of cuts made on track candidates in the CMS experiment's track trigger algorithm*

## SELECTED PUBLICATIONS

---

Only includes papers in which I made a significant contribution in reverse chronological order. A complete list can be found [here](#).

1. V. Bresò-Pla et al. "Explicit or Implicit? Encoding Physics at the Precision Frontier". *Preprint: 2603.08802*.

2. A. Petitjean et al. “Generative unfolding of jets and their substructure”. *Preprint*: [2510.19906](#).
3. A. Shmakov et al. “Full event particle-level unfolding with variable-length latent variational diffusion”. *SciPost Phys.*, **18**(4):117, 2025.
4. N. Huetsch et al. “The landscape of unfolding with machine learning”. *SciPost Phys.*, **18**(2):070, 2025.
5. ATLAS Collaboration. “Accuracy versus precision in boosted top tagging with the ATLAS detector”. *JINST*, **19**(08):P08018, 2024.
6. A. Shmakov et al. “End-to-end latent variational diffusion models for inverse problems in high energy physics”. *NeurIPS proceedings*, 2023.
7. K. Greif and K. Lannon. “Physics Inspired Deep Neural Networks for Top Quark Reconstruction”. *EPJ Web Conf.*, **245**:6029, 2020.

## PRESENTATIONS

---

|                  |                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------|
| <b>Jul. 2025</b> | “Forward folding versus unfolding in the age of ML”<br>ML4Jets 2025                                     |
| <b>Nov. 2024</b> | “Full Event Particle-Level Unfolding with Variable Length Variational Latent Diffusion”<br>ML4Jets 2024 |
| <b>Nov. 2023</b> | “Systematic Effects in Jet Tagging Performance for the ATLAS Detector”<br>ML4Jets 2023                  |
| <b>Nov. 2023</b> | “End-to-End Latent Variational Diffusion Models for Unfolding LHC Events”<br>ML4Jets 2023               |
| <b>Nov. 2022</b> | “Constituent-Based Top-Quark Tagging with the ATLAS Detector”<br>ML4Jets 2022                           |
| <b>Nov. 2019</b> | “Physics Inspired Deep Neural Networks for Top Quark Reconstruction”<br>CHEP 2019                       |

## APPOINTMENTS AND MEMBERSHIPS

---

|                                                                                                                                                                                                                                    |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>ATLAS Early Career Scientist Board</b>                                                                                                                                                                                          | 2024 - Present |
| <ul style="list-style-type: none"> <li>• Advocate for the interests of early career scientists within the ATLAS collaboration</li> <li>• Organize educational and career development events for early career scientists</li> </ul> |                |
| <b>Phi Beta Kappa Society</b>                                                                                                                                                                                                      | 2020 - Present |
| <b>Sigma Pi Sigma Society</b>                                                                                                                                                                                                      | 2020 - Present |

## AWARDS

---

**Outstanding Physics Major** Physics Department, University of Notre Dame, 2020

**Paul Chagnon Award in Physics** Physics Department, University of Notre Dame, 2020

*Recognizing service to the department and fellow students*

**Edward J. Cronin Award** Program of Liberal Studies, University of Notre Dame, 2020

*Awarded for the finest piece of written work submitted in an academic year*

**Hichwa Fellowship** Physics Department, University of Notre Dame, 2019

## OUTREACH

---

**ATLAS Virtual Visit Tour Guide** 2022 - 2025

*Remote visits to the ATLAS experiment for students from around the world.*

**UCI Physics and Astronomy Blog** 2020 - 2022

*Publish short, accessible [descriptions](#) of new research from the UCI physics department.*

## TEACHING

---

- Physics 3A (Mechanics) TA, Fall 2021
- Physics 3LB (Electricity and Magnetism Lab) TA, Spring 2021
- Physics 7LC (Mechanics Lab) TA, Winter 2021
- Physics 2 (Mathematical Methods in Physics) TA, Fall 2021
- Computational Methods in Physics TA, Spring 2020

## REFERENCES

---

Daniel Whiteson: [daniel@uci.edu](mailto:daniel@uci.edu)

Ben Nachman: [nachman@stanford.edu](mailto:nachman@stanford.edu)

Kevin Lannon: [klannon@nd.edu](mailto:klannon@nd.edu)